

Push 2 / Nuendo Bridge

User Guide & Installation Manual — Version 1.0.4

Compatible with Nuendo 14+ and Cubase 14+ — macOS and Windows

This guide covers installation, configuration, and use of the Push 2 / Nuendo Bridge.

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1. System Requirements

- Ableton Push 2 (USB), Steinberg Nuendo 14+ or Cubase 14+, macOS 11+ or Windows 10/11
- **Python 3.11.9 maximum** (3.9–3.11.9). Newer 3.12/3.13 builds are not yet supported by the audio/MIDI dependencies.
- libusb (macOS: `brew install libusb`)
- push2-python must be installed from source:
`pip install git+https://github.com/ffont/push2-python.git`
- Plugin Mapper (optional): `pip install fastapi uvicorn pedalboard`

Nuendo 15+ features: Plugin Browser, DirectAccess insert control and the Channel Strip extended-parameter / EQ-curve pages require API 1.3. All other features work with Nuendo 14+.

2. Installation — macOS

Option A: Standalone App (Recommended)

- Step 1: Install Homebrew (if not already installed) — open Terminal and run:
`/bin/bash -c "$(curl -fsSL https://raw.githubusercontent.com/Homebrew/install/HEAD/install.sh)"`
See <https://brew.sh> for details.
- Step 2: Install libusb — `brew install libusb`
- Step 3: Copy Push2 Nuendo Bridge.app to /Applications
- Step 4: The app is **not code-signed**, so macOS Gatekeeper blocks it on first launch. Open Terminal and remove the quarantine flag (match the version in the .app name to your copy):

```
xattr -dr com.apple.quarantine "/Applications/Push2 Nuendo Bridge_v1.0.4.app"
```
- Step 5: Copy Ableton_Push2.js to:
~/Documents/Steinberg/Nuendo/MIDI Remote/Driver Scripts/Local/Ableton/Push2/
Create the Ableton/Push2 folder if it doesn't exist.
- Step 6: Double-click the app. A P2 icon appears in the menu bar.

3. Installation — Windows

Windows now ships as a **standalone .exe** — no Python, pip or command line required. See the separate Windows Installation Guide for the step-by-step walkthrough. Key points:

- Download **Push2NuendoBridge-vX.Y.Z-Windows.zip** and unzip it anywhere (e.g. Documents). It contains the .exe, Ableton_Push2.js and the PDF guides.
- Install **loopMIDI** and create the four ports: NuendoBridge In, NuendoBridge Out, BridgeNotes, BridgeNotes In. Set it to start with Windows.
- Install the **WinUSB driver for the Push 2 with Zadig** (one time) so the bridge can talk to the device over USB. **Critical:** target the Push 2's **display/bulk interface** (USB ID 2982:1967) — **NOT** the MIDI interface, or you will break MIDI on the Push. Full step-by-step in the Windows Installation Guide.

Note: the bridge and Ableton Live share the same Push 2 USB endpoint, so they cannot both control the Push at the same time — close one to use the other. On recent Live versions (12.x tested) the WinUSB driver Zadig installs works fine for Live too, so no driver swap is needed in normal use. The Windows Installation Guide covers how to restore Ableton's original driver if you ever need to (for older Live setups). macOS is not affected.

- **Double-click the .exe** to run the bridge — a console window shows the status and a clickable Plugin Mapper link (<http://localhost:8100>).

No Python install is needed: the interpreter, all dependencies and the libusb runtime are bundled inside the .exe.

- Copy **Ableton_Push2.js** to the MIDI Remote scripts folder, creating the sub-folders if needed:

```
Nuendo:  
C:\Users\<user>\Documents\Steinberg\Nuendo\MIDI Remote\Driver  
Scripts\Local\Ableton\Push2\  
  
Cubase:  
C:\Users\<user>\Documents\Steinberg\Cubase\MIDI Remote\Driver  
Scripts\Local\Ableton\Push2\
```

Replace <user> with your Windows user name. The Local\Ableton\Push2 folders usually do not exist yet — create them.

4. First Launch

Connect Push 2 via USB, launch the bridge, open Nuendo. Menu bar icon: P2 ✓ = connected, P2 ■ = waiting. Click for Start/Stop, Show Logs, Plugin Mapper, Start at Login, Quit.

5. Nuendo Configuration

Automatic setup: Nuendo usually detects the script and assigns its ports on its own.

If the script does not load automatically

- Open any project.
- Studio → MIDI Remote Manager → **Add MIDI Controller Surface**.
- Vendor = **Ableton**, Model = **Push2**.
- MIDI Input = **NuendoBridge Out**, MIDI Output = **NuendoBridge In** (the ports carry the same names as listed here).

Note Input

Set the instrument track's MIDI input to **BridgeNotes** (enable it in MIDI Port Setup if hidden).

Important — avoid doubled notes

- In Studio Setup → MIDI Port Setup, **uncheck “In ‘All MIDI Inputs’” for both Push 2 ports** — “Ableton Push 2” (Live Port) and “Ableton Push 2 User Port”.

If left checked, every pad you play is received twice: once via the bridge's BridgeNotes port and once directly from the Push 2 hardware. Excluding the raw Push 2 ports is standard practice for any control-surface bridge

and is permanent once set.

6. Using the Bridge

8 vertical zones on the Push 2 screen. Left/Right = bank by 8 tracks. Shift+Left/Right = nudge by 1 track (Volume/Pan modes).

Button	Mode	Encoders
Mix	Volume	8 track volumes
Shift+Mix	Track	Vol+Pan+Sends
Clip	Sends	8 sends
Shift+Clip	Pan	8 track pans
Device	Quick Controls	8 QC
Browse	Inserts	Plugin params
Add Device	Plugin Browser	Browse/load plugins

Lower row: Mute toggles Mute/Monitor. Solo toggles Solo/Rec. Shift+Mute/Solo clears all.

7. Mixer Modes

Volume Mode (Mix button)

Shows 8 tracks with volume bars, VU meters, and peak clip indicators. Turn an encoder to adjust volume.

Upper row buttons (above the display):

- Short press = select track
- Shift + press = clear peak clip for that track
- Long press (1 second) = open instrument UI
- Double press = open Edit Channel Settings

Lower row buttons (below the display):

- Toggle Mute, Solo, Monitor, or Record Arm on the corresponding track
- Press the Mute button to switch between Mute and Monitor mode
- Press the Solo button to switch between Solo and Record Arm mode
- Long press (0.5s) in Mute mode = toggle Monitor on that track
- Long press (0.5s) in Solo mode = toggle Record Arm on that track
- Shift+Mute = clear all mutes (or all monitors in Monitor mode)
- Shift+Solo = clear all solos (or all rec arms in Rec mode)

Pan Mode (Shift+Clip)

Shows 8 tracks with pan position indicators. Turn an encoder to adjust pan.

Track Mode (Shift+Mix)

Combined mode for the selected track: Enc1=Volume, Enc2=Pan, Enc3-8=Sends 1-6.

8. Sends Mode

Press Clip. Encoders = send levels. Upper row = On/Off. Lower row = Pre/Post. Left/Right = navigate tracks.

9. Inserts Mode

List View: Short press upper = params. Long press = open UI. Shift+upper = params+UI. Lower = bypass. Shift+Left/Right = slots 1-8 / 9-16.

Parameter View: Encoders = 8 params. Left/Right = bank/page nav. Lower 1=UI, 2=bypass, 3=deactivate. If mapped: ★ MAPPED indicator.

10. Plugin Browser

Requires Nuendo 15+ (MIDI Remote API 1.3)

Press Add Device to open the browser.

Phase 1 — Slot Selection

The screen shows the insert slots of the selected track.

- Upper row buttons (above the display) = select the target slot
- Left/Right = switch between slots 1-8 and 9-16
- Lower row button 8 = Cancel

Phase 2 — Plugin List

- Encoder 1 = page scroll (8 at a time)
- Encoder 2 = fine scroll (highlight moves within page)
- Upper row = load plugin from that column
- Lower row button 1 = cycle to next collection
- Lower row button 8 / Browse = back / cancel

After loading, the bridge returns to Inserts mode and rescans slots automatically.

11. Quick Controls (Device Mode)

Press Device. 8 Quick Controls of the selected track. Left/Right = navigate tracks.

12. Transport Controls

Button	Function	LED
Play	Playback	White/Green/Purple
Record	Recording	Orange/Blink Red/Solid Red
Fixed Length	Cycle/Loop	
Automate	Automation cycle	White/Green/Red
Metronome	Metronome	LED reflects state

13. Automation

Automate button cycles: Off → Read → Read+Write → Write → Off.

14. Touchstrip

Shift+Layout to cycle: Pitch Bend, Mod Wheel (CC 1), Volume.

15. Note Input

64 pads: Chromatic (default) or Drum (Layout). Scale, Accent, Note Repeat available. Set instrument input to BridgeNotes.

Scale selector (Scale button): pick a musical scale or the **Piano** layout. Piano mode arranges the pads like a piano keyboard — white keys on the lower row of each octave pair, black keys on the upper row, four octaves stacked bottom-to-top, every C in purple. The root is locked to C in Piano mode; use Octave Up/Down to move the whole four-octave block.

16. Control Room

Press User. 4 pages: Main, Phones, Cues, Sources. Master encoder = Main level. User+Master = Phones.

17. Channel Strip

Extended-parameter and EQ-curve control require Nuendo 15+ (API 1.3).

From the Mixer view, press **Mix again** to enter Channel Strip mode. The overview shows the six strip sections as Cubase-coloured banners (Gate, Comp, EQ, Tools, Sat, Limiter) plus PreFilter Phase and PreGain.

Important: a strip module can only be controlled once it has been **activated manually in Nuendo**. Open the channel's strip in the Nuendo mixer and load/enable each module (Gate, Comp, EQ, Tools, Sat, Limiter) you want to use. An empty or disabled strip slot is not exposed to the Push 2 — its banner and encoders stay inactive until you turn the module on in Nuendo.

Overview

- Each banner shows the section colour (dimmed when bypassed) and the loaded plugin variant.
- Lower row 1-6 = Bypass each section (pill/LED amber when bypassed).
- Lower row 7 = PreFilter Phase On/Off. Lower row 8 = Edit Channel Settings (white when the window is open).
- Upper row 1-6 = drill into that module's sub-page.

Module Sub-pages

- 8 encoders control that module's parameters. Parameters beyond Cubase's bank zone are exposed via DirectAccess (e.g. VintageCompressor Ratio & Mix, Tube Compressor Character, DeEsser side-chain filter, Maximizer Release/Recover).
- Lower row = the module's binary toggles (Auto Release, Solo, Diff, etc.) — LEDs follow the on-screen pills.
- Upper row layout: 1 = module name (inactive), 2 = Bypass, 3-5 = variant chips, 8 = back to overview.

Variant chips show the currently loaded plugin. Switching the variant must be done in Nuendo's strip-slot menu — the MIDI Remote API does not allow changing strip-slot plugins from a controller; the chip mirrors the active variant in real time.

18. Channel EQ Page

Drill into the EQ section for an EQ-Eight-style page with a live frequency-response curve.

Encoder	Function
1	Band selector (1-4) — turn to pick the active band
2	Filter Type of the selected band (bands 2-3 are fixed Peak)
3	Frequency of the selected band
4	Q of the selected band
5	Gain of the selected band
6	PreFilter Low-Cut frequency

7	PreFilter High-Cut frequency
8	PreGain

- Lower row 1 = selected band On/Off. Lower 6 = LC On, Lower 7 = HC On, Lower 8 = PreFilter bypass.
- Upper row 2 = EQ section bypass (amber when bypassed). Upper 8 = back.
- The curve combines the 4 EQ bands + PreFilter HC/LC. Numbered circles mark each band at its (freq, gain) point — white when the band is on, grey when off. The curve and markers update live, whether you change a parameter from the Push or directly in Nuendo.

19. Setup Page

Press Setup. Tabs: MIDI Ctrl (aftertouch), Vel Curve (5 presets), CC Mode (Absolute/Pick-up), About.

CC Mode

- **Absolute** (default): encoder value is sent immediately. May cause parameter jumps if the encoder position differs from the current value in Nuendo.
- **Pick-up**: encoder does not send values until the user reverses the direction of rotation. This prevents jumps but requires a direction change to engage.

***Important:** Nuendo does not send the current CC parameter value back to the Push 2. Because of this limitation, the bridge cannot know the actual value in Nuendo and cannot implement a true pick-up (where the encoder catches up to the existing value). Instead, pick-up mode uses direction-change detection: turn the encoder in one direction (nothing happens in Nuendo), then reverse — the encoder engages from that point on.*

20. MIDI CC Controller

Shift+Note. 8 assignable CC faders on BridgeNotes port. Upper row = edit CC number. Lower row = toggle 0/127. Defaults: CC 1,2,7,8,10,11,64,65.

21. Plugin Mapper

Create custom parameter mappings for VST3 plugins. Access via menu bar (Plugin Mapper) or <http://localhost:8100>. Mappings saved in **~/push2bridge/mappings/**. See the separate Plugin Mapper Guide for details.

Requires: `pip install fastapi uvicorn pedalboard`. The bridge works without these — the Mapper is optional.

22. Button Reference

Button	Action	Shift + Button
Mix	Volume mode (press again: Channel Strip)	Track mode
Clip	Sends mode	Pan mode
Device	Quick Controls	
Browse	Inserts mode	
Add Device	Plugin Browser	
Note	MIDI note pads	MIDI CC controller
Setup	Setup page	
Left/Right	Bank nav (8)	Nudge (1) in Vol/Pan
Play	Playback	

Record	Record toggle	
Mute	Mute/Monitor	Clear all
Solo	Solo/Rec arm	Clear all
User	Control Room	
Layout	Drum/Chromatic	Touchstrip cycle
Upper (dbl / Shift)	Edit Channel Settings	
Scale	Scale selector	
Accent	Fixed velocity	
Repeat	Auto-repeat	

23. MIDI Channel Allocation

Channel	Usage
1 (0xB0)	Mixer (volume, pan, transport, selection, VU)
2 (0xB1)	Insert plugin parameters
3 (0xB2)	Send levels (selected track)
4 (0xB3)	Insert bypass toggles
5 (0xB4)	Quick Controls
6 (0xB5)	Control Room
7 (0xB6)	Bank zone sends
8 (0xB7)	DirectAccess commands (browser, bypass, edit, strip slot, param-flip, variant)
9 (0xB8)	DirectAccess encoder control (mapped params, strip slots)
16 (0xBF)	Heartbeat

24. Troubleshooting

- **Push 2 not found:** check USB, libusb, close Ableton Live.
- **push2-python error:** install it from source — `pip install git+https://github.com/ffont/push2-python.git`. Use Python 3.11.9 or lower.
- **No Nuendo connection:** check bridge is running; add the surface manually (MIDI Remote Manager → Add, Vendor Ableton / Model Push2).
- **Pads play doubled notes:** uncheck “In ‘All MIDI Inputs’” for the Push 2 Live Port and User Port (Studio Setup → MIDI Port Setup).
- **Channel Strip / EQ curve not working:** requires Nuendo 15+ (API 1.3). Drill into a section once to trigger DA strip exploration.
- **Plugin Mapper not loading:** pip install fastapi uvicorn pedalboard.
- **Track names not loading:** navigate Left/Right to refresh.

Logs: macOS — ~/Library/Logs/Push2NuendoBridge.log. Windows — terminal output.

25. Support

Push 2 / Nuendo Bridge is donationware.

- GitHub: <https://github.com/mbourque-mix/Push2Nuendo-Bridge>
- Buy Me A Coffee: <https://buymeacoffee.com/mbourque>
- Issues: <https://github.com/mbourque-mix/Push2Nuendo-Bridge/issues>

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